

1. What is a CLI?

CLI (Command-Line Interface) is a text-based interface used to configure and manage Cisco devices.

- Configure routers and switches
- Faster for network administrators
- Opposite of GUI (Graphical User Interface)

2. Connecting to a Cisco Device

Initial Access: Use the Console Port for first-time setup.

Default Console Settings:

- Speed: 9600 baud
- Data Bits: 8
- Stop Bits: 1
- Parity: None
- Flow Control: None

3. Cisco IOS CLI Modes

Router> → User EXEC Mode

Router# → Privileged EXEC Mode

Router(config)# → Global Configuration Mode

Router(config-if)# → Interface Configuration Mode

4. Navigation Commands

- enable → Enter Privileged EXEC
- disable → Return to User EXEC
- configure terminal (conf t)
- exit → One level up
- end / Ctrl+Z → Return to Privileged EXEC
- ? → Help
- Tab → Auto-complete

5. Password Configuration

enable password → Plain text (weak) **enable secret** →

Encrypted (recommended) **service password-encryption**

→ Encrypts line passwords **Exam Tip:** Always prefer

enable secret.

6. Configuration Files

running-config → Active configuration in RAM **startup-config** → Saved configuration in NVRAM

7. Save Configuration

- write
- write memory
- copy running-config startup-config (**Recommended**)

8. Useful Show Commands

- show running-config
- show startup-config
- show version

9. The 'no' Command

Removes existing configuration.

Example: **no service password-encryption**

Summarize

CLI Access: Console → PuTTY → 9600,8,N,1

Modes: Router> → enable → Router# → conf t → Router(config)#

Save: copy run start

Security: enable secret

Help: ? and Tab

Remove Commands: no <command>